

A PROJECT REPORT ON

**A Dual-Phase Framework for Intelligent File
Categorization and Real-Time Anomaly
Detection in File Operation Behaviour**

**SUBMITTED TO THE SAVITRIBAI PHULE UNIVERSITY, PUNE
IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF THE DEGREE**

OF

BACHELOR OF ENGINEERING (Computer Engineering)

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2024-2025**



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ACKNOWLEDGEMENT

It gives us great pleasure in presenting the project report on ‘A Dual-Phase Framework for Intelligent File Categorization and Real-Time Anomaly Detection in File Operation Behaviors’.

*We would like to express our sincere gratitude to our mentor and project guide, **Prof. Ashwini Bunde**, for providing the problem statement that forms the foundation of this project. Their guidance and support have been invaluable throughout the research and development phases of this project. We are deeply thankful for their continuous encouragement, insightful feedback, and valuable suggestions that greatly contributed to shaping the direction of this work.*

*We are also thankful to our Head of the Department, **Dr. G.V. Kale**, for their indispensable support and timely feedback throughout the project.*

Additionally, we would like to acknowledge the collaboration and support from our peers, which significantly enhanced the overall completion of the project.

This project would not have been possible without the resources and infrastructure provided by the institution, and the feedback-driven model improvement system incorporated into the project has been a key factor in ensuring its success.

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ABSTRACT

*This project, **A Dual-Phase Framework for Intelligent File Categorization and Real-Time Anomaly Detection in File Operation Behaviors**, addresses two critical challenges in modern file management systems: accurate categorization of files and proactive detection of anomalous file activities. The first phase focuses on developing a robust and scalable classification system that predicts a file's category based on metadata such as file extension, MIME type, and description. By leveraging supervised machine learning models like Decision Trees and Random Forests, supplemented with MIME type verification through the magic library, the system ensures high accuracy even in the presence of misleading or altered extensions. A dynamic learning mechanism retrains the model with unseen extensions after reaching a threshold, while a custom hash-based data structure is also implemented to enhance classification reliability.*

The second phase introduces a real-time anomaly detection system that logs and monitors file operations—such as insertions, updates, renames, and deletions—using tools like `os`, `watchdog`, and `ZODB` for persistent, structured storage. Multiple detection models, including Autoencoder, Isolation Forest, Local Outlier Factor, One-Class SVM, and River, are combined through ensemble aggregation for reliable anomaly detection. Clustering techniques like DBSCAN group similar anomalies, which are then interpreted using Ollama Llama3 language model to provide human-readable, context-aware explanations. Parallel processing optimizations ensure fast response times, while a React-based dashboard visualizes anomalies and notifies users based on severity levels.

In conclusion, this project presents a comprehensive solution that not only streamlines document classification but also strengthens operational security through real-time anomaly detection and explainability—enhancing the intelligence, transparency, and resilience of file management systems.

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CHAPTER 1

INTRODUCTION

1.1 Overview

In an increasingly digital world, efficient file management is critical for maintaining the integrity, accessibility, and security of information systems. As organizations handle vast and diverse file types daily, the challenges of accurately categorizing files and detecting irregular file operations have grown significantly. Misclassification due to tampered extensions or undetected anomalies in file behavior can lead to data mismanagement, loss, or even security vulnerabilities. This project addresses these challenges through a dual-focus system: intelligent file categorization and behavioral anomaly detection. The first component seeks to improve how files are classified—not just by extension, but by learning underlying patterns from curated datasets, ensuring better accuracy even when file metadata is misleading or incomplete. The second component monitors operational behaviors of files, such as creation, updates, renaming, and deletion, to identify deviations from expected usage patterns. By integrating learning-based classification with behavior monitoring, the system aims to support real-time, adaptive, and context-aware file management. Such an approach holds value in various sectors, including enterprise data governance, research data handling, and cybersecurity, where automation, reliability, and transparency in file handling processes are paramount.

1.2 Motivation

As digital ecosystems scale, the volume and diversity of file types handled by organizations have grown exponentially. Traditional methods of file management—often reliant on manual classification or simplistic extension-based sorting—are no longer sufficient in environments where precision, speed, and security are paramount. Misclassification of files due to tampered or ambiguous extensions can lead to chaotic data organization, reduced productivity, and increased risk of operational failures. Simultaneously, file operations such as sudden deletions, frequent modifications, or unauthorized renaming may serve as early indicators of system misuse or potential security breaches. Yet, these subtle behavioral anomalies often go unnoticed in static monitoring systems. This project is driven by the need to bridge these gaps through automation and intelligence. By developing a robust classification system that learns from contextual metadata and an anomaly detection mechanism that continuously tracks file behavior, the aim is to create a more secure, efficient, and self-adaptive file management framework. The ultimate goal is to empower organizations with tools that not only streamline data handling but

also proactively safeguard digital assets against mismanagement and threats.

1.3 Problem Definition and Objectives

The project addresses two interrelated challenges in modern file management systems:

1.3.1 File Extension-Based Categorization

In large-scale data environments, organizing files by type is crucial for efficient access, storage optimization, and information retrieval. However, categorizing files solely based on their extensions is inherently error-prone, as extensions can be misleading, tampered with, or ambiguous.

The first problem, therefore, is to develop an intelligent classification system that accurately predicts a file's category—such as *Image Files*, *Executables*, or *Compressed Archives*—based on its extension and associated metadata. The goal is to automate and enhance the reliability of file organization within complex repositories.

1.3.2 Behavioral Anomaly Detection in File Operations

While file categorization streamlines organization, maintaining system integrity requires vigilant monitoring of file activity. Sudden or unusual patterns in file operations—like a spike in deletions, unauthorized renaming, or abnormal update frequencies—may indicate threats such as unauthorized access, user errors, or malware activity.

The second problem lies in creating a real-time anomaly detection system that learns from historical file operation data to recognize normal usage patterns and flag deviations that may pose risks to security or data integrity.

Together, these problems define the scope of a comprehensive system that not only classifies files intelligently but also safeguards the environment through continuous behavior analysis.

1.4 Project Scope & Limitations

1.4.1 Project Scope

The proposed project, *A Dual-Phase Framework for Intelligent File Categorization and Real-Time Anomaly Detection in File Operation Behaviors*, focuses on building an intelligent system that addresses the growing need for accurate file classification and secure, explainable anomaly detection. The framework is structured into two primary phases:

- **Phase I – File Categorization:** Utilizes supervised machine learning models alongside MIME-type validation and a hash-based structure to classify files into categories such as Documents, Audio, Images, and Source Code. It supports both file-based and directory-based classification and dynamically learns from unknown extensions over time.
- **Phase II – Anomaly Detection:** Monitors and logs file operations (insert, update, rename, delete) and employs an ensemble of anomaly detection models (Autoencoder, LOF, Isolation Forest, One-Class SVM, etc.). These are aggregated to detect abnormal behaviors and their severity. Further, a locally deployed LLM provides human-readable explanations of anomalies to enhance transparency and user trust.

The system is designed to operate across distributed environments with support for offline usage and efficient storage using ZODB, ensuring long-term scalability and performance. The insights are displayed in real-time through a web-based dashboard that issues alerts based on detected anomalies.

1.4.2 Limitations in Existing Systems

Prior research and industrial systems in file classification and anomaly detection suffer from multiple limitations that this project aims to address. These include:

- **Scalability Bottlenecks:** Most systems struggle to handle large volumes of file operation data efficiently. Our implementation resolves this with ZODB—a scalable object-oriented database supporting offline access and distributed systems.

- **Ambiguity in Classification:** Conventional models often rely solely on file extensions, leading to misclassifications. This project uses MIME type analysis to verify file content and reduce errors from tampered extensions.
- **Lack of Adaptability:** Existing solutions attempt to generalize user behavior without accounting for system-specific patterns. In contrast, our approach clusters and categorizes events without assuming uniform behavior, increasing detection accuracy.
- **High Resource Utilization:** Cloud-based or continuously learning systems tend to be resource-heavy. Our framework is optimized for local processing, enabling faster response time and reduced system load.
- **Absence of Explainability:** Traditional anomaly detection systems often lack transparency in decision-making. Our integration of an on-premise LLM offers interpretable, human-friendly outputs for each flagged anomaly.

This project thus not only defines a robust, modular system but also systematically mitigates key shortcomings found in existing solutions in the domains of file intelligence and operational security.

1.5 Methodologies of Problem Solving

The proposed system is developed using a structured, dual-phase methodology that addresses two core challenges: intelligent file categorization and real-time anomaly detection. The methodologies followed in both phases are as follows:

1.5.1 Phase I: Intelligent File Categorization

This phase aims to automatically classify unknown files into appropriate categories (e.g., Documents, Audio, Images, Source Code) using a hybrid approach involving machine learning, MIME-type verification, and hash-based matching.

- **Data Collection:** Over 20,000 file extensions were scraped from online repositories like fileinfo.com. Each entry included attributes such as file extension, category, description, and MIME type.

- **Preprocessing:** Duplicate and inconsistent entries were cleaned. Ambiguous categories were consolidated to improve model reliability.
- **Supervised Learning Models:** Decision Tree and Random Forest classifiers were trained using file extension metadata to predict file categories.
- **MIME Type Verification:** The ‘python-magic’ library was used to validate file types by checking actual content, minimizing extension spoofing.
- **Fallback Hash-Based Matching:** For higher accuracy, a hash-based method was implemented as an additional classification layer, ensuring correctness in edge cases.
- **Dynamic Learning:** Unknown file types are temporarily cached. Once a threshold is reached, they are validated, added to the dataset, and the model is retrained.

1.5.2 Phase II: Real-Time Anomaly Detection

This phase focuses on detecting suspicious file operation behaviors (insert, update, rename, delete) using ensemble-based unsupervised learning combined with local explainability.

- **Logging System:** A custom event-logger tracks file operations in real-time using the ‘os’ and ‘watchdog’ libraries. Logs are stored in a scalable ZODB database (.fs file).
- **Feature Extraction:** Extracted features include file path, operation type, time, bytes modified, and old/new names. These are formatted for anomaly model input.
- **Anomaly Detection Models:** Multiple models (Autoencoder, Isolation Forest, One-Class SVM, LOF, and River Model) are applied in parallel, and their outputs are aggregated using ensemble voting to reduce false positives.
- **Severity Classification:** Anomalies are tagged with severity levels (Low, Moderate, High) based on feature deviation and model consensus.

- **Clustering and Explainability:** Detected anomalies are clustered using DBSCAN. A locally hosted LLM then generates natural language explanations for each cluster, enhancing transparency for end-users.
- **Parallel Optimization:** Threaded processing and batch evaluation optimize runtime performance across large datasets.

1.5.3 Deployment Architecture

The system uses a modular Flask backend integrated with a React-based frontend dashboard for real-time visualization and alert management. ZODB ensures efficient offline data handling and local persistence.

This methodological pipeline ensures that the system remains explainable, adaptive, and scalable across environments with minimal human intervention.

CHAPTER 2
LITERATURE SURVEY

2.1 File Categorization Based on File Extensions

2.1.1 Traditional Approaches to File Categorization

Early file categorization systems were primarily rule-based, mapping file extensions to predefined categories. **Rubrik (2020)** described how simple rule-based systems classified files by their extensions (e.g., ".docx" as "Word Document" or ".jpg" as "Image File"). These approaches, though straightforward, struggled with scalability when handling new or ambiguous file types. In contrast, **Author B et al. (2019)** developed a heuristic-based categorization model that performed well in controlled environments but lacked the adaptability required for modern, diverse datasets.

2.1.2 Machine Learning for File Categorization

The transition from rule-based systems to machine learning significantly improved the accuracy of file categorization. **Springer et al. (2020)** demonstrated the effectiveness of **Random Forests** in classifying documents based on file extensions. Their model was able to handle overlapping extensions (e.g., ".txt" for text files, logs, or scripts) with an accuracy of 92%. Similarly, **Author Z et al. (2018)** employed **Support Vector Machines (SVMs)** for document classification, focusing on cases where file extensions were ambiguous. Their model achieved high accuracy but required careful hyperparameter tuning to prevent overfitting. A further enhancement came with the integration of **decision trees**. **ResearchGate (2020)** explored how decision trees provided transparency and interpretability in document classification. However, their tendency to overfit remained a challenge, particularly when handling large datasets with multiple file categories.

2.1.3 Neural Networks and Deep Learning Approaches

Deep learning approaches, particularly **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)**, have proven especially effective for large-scale file categorization tasks. **Springer et al. (2021)** introduced a CNN model that significantly improved classification accuracy in complex datasets, achieving over 95% accuracy when applied to multimedia file types such as ".mp4" and ".avi". **Author Y et al. (2020)** used RNNs for sequential document categorization, focusing on file extensions with multiple components (e.g., ".tar.gz"). Their approach handled

sequential patterns well but required substantial training data for optimal performance.

2.1.4 Effectiveness of Different Models

SIFT: Sifting File Types—Application of Explainable AI in Cyber Forensics highlights the application of Explainable AI (XAI) in file classification, providing transparency in model decisions. This paper compares the explainability of machine learning models, which is a crucial aspect when evaluating the effectiveness of classification models for file categorization.

2.1.5 Feature Selection and Data Preprocessing

Feature selection plays a pivotal role in improving the accuracy of file categorization models by eliminating irrelevant data and focusing on essential attributes. Common techniques such as **Principal Component Analysis** (PCA) and **Linear Discriminant Analysis** (LDA) are frequently employed to reduce dimensionality, enabling models to operate efficiently on large datasets. In file categorization, preprocessing steps such as data normalization and cleaning are critical, particularly when dealing with heterogeneous data formats. Techniques like data augmentation can also be leveraged to handle imbalanced datasets, which are common in real-world file management systems. **Nilesecure (2022)** highlighted the importance of cleaning and normalizing file extension data before inputting it into machine learning models. Inconsistent capitalization (e.g., ".jpg" vs ".JPG") and obsolete extensions were identified and removed to improve model accuracy. Principal Component Analysis (PCA) and dimensionality reduction techniques were employed by **Author A et al. (2015)** to streamline the feature set, enhancing model efficiency and reducing computation time without compromising accuracy.

2.2 Anomaly Detection in Categorized File Behavior

2.2.1 Time-Series Analysis for Anomaly Detection

Anomaly detection is essential for identifying irregular behavior in file systems, such as unusual patterns in file additions, updates, or deletions. **Trendz**

(2022) applied time-series models to detect abnormal spikes in file operations, using **ARIMA** to predict regular behavior and flag deviations. **Author D et al. (2016)** further explored Exponential Smoothing as a method to capture subtle anomalies, reducing false positives in environments with fluctuating file activity. Their study showed that smoothing techniques provided more reliable anomaly detection in volatile environments.

2.2.2 Clustering for Unsupervised Anomaly Detection

Clustering algorithms, such as **k-means** and **DBSCAN**, are frequently applied in unsupervised anomaly detection, helping to identify abnormal file access patterns or unusual file behaviors in real-time systems. For instance, in environments with high activity, DBSCAN's ability to detect arbitrary shaped clusters is particularly useful for identifying anomalies in dynamic datasets. On the other hand, k-means is often preferred for systems with moderate to low activity levels due to its computational efficiency. Studies have shown that combining clustering algorithms with other techniques like time-series analysis can significantly improve anomaly detection accuracy. Unsupervised learning techniques, such as k-means and DBSCAN, have been employed to detect anomalies by clustering file behavior patterns. **Springer et al. (2020)** applied DBSCAN to cluster normal file operations and identify outliers, with particular success in detecting unauthorized deletions in file management systems. **Author F et al. (2019)** integrated clustering with time-series analysis to further enhance anomaly detection, reducing false positives and improving the system's ability to detect low-frequency anomalies, such as sporadic data corruption.

2.2.3 Autoencoders for Anomaly Detection

Autoencoders have become increasingly popular in unsupervised anomaly detection tasks. **IEEE (2021)** proposed an autoencoder-based model that learned normal file behavior by compressing high-dimensional data, then flagging deviations as potential anomalies. This approach worked particularly well in high-dimensional datasets, such as large-scale file systems in enterprise environments. Additionally, **ResearchGate (2021)** applied autoencoders to log data from physical access control systems, achieving a 15% reduction in false positives and enabling real-time detection of abnormal user activities.

2.2.4 Generative Adversarial Networks (GANs)

Adversarial Networks and Machine Learning for File Classification proposes a semi-supervised GAN model to classify files even when file extensions or headers are obfuscated. This method achieved 97.6% accuracy across 11 file types, making it a strong candidate for addressing file obfuscation challenges. This approach improves classification in anomaly detection and file categorization by handling misleading file types.

2.2.5 Content-Based Detection Using PCA and Neural Networks

A New Approach to **Content-Based File Type Detection** introduces a PCA-based method combined with neural networks to detect file types based on content, ensuring both accuracy and speed. This model aids in clustering and categorizing file extensions with more precision, helping to overcome ambiguities where file extensions alone are insufficient.

2.3 Feature Engineering and Data Preprocessing

In file categorization, content-based feature engineering techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) are widely used to extract meaningful patterns from text-based files. Combining clustering techniques with neural networks, such as using autoencoders for unsupervised feature extraction, enhances the accuracy of file classification. In real-time systems, feature engineering should be designed to minimize computational overhead while maximizing the discriminatory power of the selected features.

2.4 Feedback Mechanisms for Model Improvement

Feedback loops are a critical component in continuously improving the accuracy of both categorization and anomaly detection models. **Rubrik (2020)** emphasized the role of user **feedback in refining categorization models**. Their system incorporated feedback from misclassified files, which was used to retrain the model periodically, increasing overall accuracy by 10%. **Author H et al. (2018)** implemented a feedback mechanism for anomaly

detection models, allowing system administrators to verify flagged anomalies. This feedback loop reduced the number of false alarms while ensuring the system remained sensitive to real threats. Integrating user feedback into file categorization and anomaly detection systems provides a dynamic mechanism for continuous model improvement. A feedback loop allows users to confirm or correct detected anomalies, which can be fed back into the system to refine model parameters. For example, systems like Rubrik incorporate feedback mechanisms that reduce false positives in anomaly detection by re-training models based on user inputs. Feedback-driven models, particularly in hybrid systems combining rule-based approaches with machine learning, have shown significant improvements in adaptability and precision.

2.5 Integration of File Categorization and Anomaly Detection

Few studies have fully integrated categorization and anomaly detection into a unified system. **Springer et al. (2021)** proposed a hybrid framework that first categorised files based on their extensions and then monitored file behaviour for anomalies. This approach, combining neural networks for categorization with time-series analysis for anomaly detection, achieved a 25% reduction in security breaches due to unauthorized file modifications. **Author I et al. (2020)** explored a similar integration, focusing on enhancing cybersecurity through the dual application of categorization and anomaly detection. Their work showed that categorizing files before anomaly detection allowed for more accurate threat identification.

CHAPTER 3

**SOFTWARE REQUIREMENTS
SPECIFICATION**

3.1 Assumptions and Dependencies

3.1.1 Assumptions

- **Availability of Consistent File Metadata**

It is assumed that critical metadata—such as file extensions, MIME types, timestamps, and other descriptors—are consistently accessible and accurate across all file operations.

- **Stable File System Structure**

The solution presumes a stable file system architecture during active monitoring. Any significant structural modifications to the file system are expected to be infrequent or pre-announced.

- **Availability of Historical Data for Training**

It is assumed that a sufficient volume of historical log data is present, enabling the training of machine learning models to accurately capture normal operational behaviors.

- **Detectable Anomaly Patterns**

The system operates under the assumption that abnormal or malicious file activities produce recognizable deviations from learned norms, allowing them to be flagged effectively as anomalies.

- **Consistency in User Behavior**

It is presumed that the behavior of individual users or user roles remains relatively consistent over time. This stability enables the creation of accurate behavioral baselines for anomaly detection.

3.1.2 Dependencies

- **Third-party Libraries and Frameworks**

The project depends on several external Python libraries and frameworks for machine learning, data processing, and file system monitoring, including but not limited to:

- Scikit-learn for traditional ML models
- TensorFlow for deep learning components
- Pandas for data manipulation
- Watchdog for file system event tracking

- **Large Language Model for Explainability**

The system's explainability module relies on access to a locally hosted large language model (e.g., Ollama's LLaMA3) or an equivalent LLM to generate human-readable explanations of detected anomalies.

- **Hardware Acceleration (GPU Support)**

For efficient training and real-time inference, particularly for deep learning tasks and high-dimensional feature extraction, the system requires access to GPU hardware acceleration.

- **Operating System Compatibility**

Development and testing have been conducted on Ubuntu 22.04 LTS. Full compatibility on other operating systems (e.g., Windows, macOS) may require additional adjustments or platform-specific handling.

- **System Permissions and Access Rights**

The anomaly detection system requires elevated file system permissions to monitor directories, access file metadata, and log operations. These permissions may vary across deployment environments based on system security policies.

3.2 Functional Requirements

3.2.1 System Feature 1: File Extension Categorization (Functional Requirement)

Input: File extension (e.g., .docx, .jpg).

Output: Predicted file category (e.g., "Word Document", "Image"). **Functional Behavior:**

- The system will process file extensions as inputs.
- It will match the input with predefined categories using a trained machine learning model.
- The categorized output will be integrated into the file management system for further actions such as organization or search indexing.

3.2.2 System Feature 2: Anomaly Detection in File Type Behavior (Functional Requirement)

Input: Real-time data on file operations (e.g., add, update, delete).

Output: Flagging of anomalies based on learned patterns.

Functional Behavior:

- The system will monitor file operations for each file category.
- A machine learning model will compare real-time data with historical patterns to identify unusual activities (e.g., excessive deletions in a category).
- Anomalies will trigger alerts for administrators to review.

3.3 External Interface Requirements

This section outlines the interfaces through which users, hardware, and external systems interact with the proposed framework for intelligent file categorization and real-time anomaly detection.

3.3.1 User Interfaces

- **File Categorization Dashboard:** A responsive and intuitive interface allows users to input directories or individual files for classification. The system then displays the predicted category, associated metadata, and MIME verification results.
- **Anomaly Detection Interface:** An administrative dashboard is provided to monitor real-time alerts, review anomaly logs, and access explanations for flagged operations. Users can provide feedback to enhance the system's adaptive capabilities.
- **Severity Visualization:** Anomalies are tagged with severity levels (Low, Moderate, High), and are represented with corresponding visual cues for quick interpretation.

3.3.2 Hardware Interfaces

- **Standard File System Integration:** The system integrates with the host machine's file system (or mounted volumes) to monitor operations such as file insertion, renaming, updates, and deletions.
- **Server Infrastructure:** The application runs on general-purpose computing hardware. No specialized hardware is required beyond typical server or desktop configurations capable of running Python-based services.

- **Optional Peripheral Storage:** For large-scale deployments, external storage or network-attached storage (NAS) may be used for persistent log archiving.

3.3.3 Software Interfaces

- **Operating System and File I/O APIs:** Utilizes system-level APIs (e.g., via `os`, `watchdog` libraries) to track file activity in real time.
- **Machine Learning Frameworks:** Core classification and anomaly detection models are implemented using libraries such as `Scikit-learn`, `TensorFlow`, and `PyTorch`, depending on the model requirements.
- **Database Integration:** File operation logs and classification records are stored in a ZODB object-oriented database to ensure offline availability and scalability.
- **Web Frameworks and UI:** The backend is built using `Flask`, while the user-facing dashboards are rendered through a `React.js` frontend, enabling dynamic data visualizations and interaction.
- **LLM Explainability API (Optional):** For explanation generation, the system optionally connects to a locally hosted or remote LLM (e.g., via Groq API), receiving natural language explanations for clustered anomalies.

3.4 Non-Functional Requirements

This section outlines the critical non-functional attributes of the system that ensure its efficiency, security, reliability, and usability in practical deployment scenarios.

3.4.1 Performance Requirements

- The file extension classification module must deliver predictions with a latency of less than 100 milliseconds to ensure seamless user experience.
- The anomaly detection pipeline should process incoming file operations in real time, generating alerts with a delay not exceeding 1–2 seconds from the event trigger.

- The system architecture should support horizontal scalability, capable of handling and logging over 10,000 file operations per minute without performance degradation, especially in enterprise environments.
- Concurrent processing and thread-level optimization must be employed to maintain responsiveness under high-volume activity.

3.4.2 Safety Requirements

- The system shall not perform automated modifications (deletion, overwrite) on user files without explicit administrative approval, preventing unintended data loss.
- Alerts generated by the anomaly detection engine must be presented in a clearly distinguishable format from regular system logs or status messages to avoid misinterpretation.
- Fail-safes should be embedded in the logging mechanism to ensure that operation logs are preserved even during unexpected system shutdowns or crashes.

3.4.3 Security Requirements

- All data—especially classified file operations and user credentials—must be handled using secure protocols (e.g., HTTPS, encryption at rest and in transit).
- Role-based access control (RBAC) should be implemented to restrict sensitive anomaly review and feedback features to authorized personnel only.
- All activity logs and anomaly reports must be tamper-proof, cryptographically signed, and securely stored to support traceability and forensic investigation.
- The system should comply with data protection and privacy regulations applicable to the deployment context (e.g., GDPR, HIPAA, or equivalent).

3.5 System Requirements

This section outlines the essential software and hardware resources required for the development, execution, and deployment of the proposed dual-phase

framework. The software stack supports model training, logging, real-time monitoring, and local LLM integration, while hardware requirements ensure smooth and scalable performance across organizational environments.

3.5.1 Software Requirements

LIST OF SOFTWARE REQUIREMENTS

Category	Requirements
Operating System	Ubuntu 22.04 LTS (primary support), Compatible with Windows 10/11, macOS
Programming Language	Python 3.8+
Key Libraries & Frameworks	Scikit-learn, TensorFlow or PyTorch, Pandas, Watchdog, DBSCAN, ZODB, Magic, Matplotlib, River
Database	ZODB (for anomaly logs)
LLM Integration	Local LLM: Ollama LLaMA3 or (on-prem deployment)
IDE/Development Tools	Visual Studio Code, PyCharm, or Jupyter Notebook
Version Control	Git with GitHub or GitLab for source code and model tracking
Containerization (Optional)	Docker (for environment isolation and deployment)

The software stack has been chosen for its compatibility, scalability, and support for local model inference, ensuring efficient development and secure deployment within restricted network environments.

3.5.2 Hardware Requirements

The following hardware specifications are recommended for optimal performance during the development, execution, and deployment phases.

LIST OF HARDWARE REQUIREMENTS

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i5 or equivalent	Intel Core i7 / AMD Ryzen 7 or higher
RAM	8 GB	16–32 GB (for improved parallel processing)
Storage	256 GB SSD	512 GB SSD or higher (for faster I/O operations)
GPU (for ML/DL)	NVIDIA GTX 1050 (2GB VRAM)	NVIDIA RTX 2060/2070 or higher (6–8GB VRAM)
Display	1080p resolution	Full HD or higher for dashboard visualization
Connectivity	Wi-Fi/Ethernet	Stable internet for updates and LLM integration

3.6 Analysis Models: SDLC Model to be applied

The Agile model is selected for this project due to its flexibility, adaptability, and emphasis on continuous improvement. This development approach is particularly well-suited for building a dynamic Anomaly Detection System that evolves with user interaction and environmental changes.

3.6.1 Rationale for Agile Selection

- **Iterative Development:** The system is developed in short, manageable sprints, allowing for incremental delivery of key features such as file categorization, real-time logging, model-based anomaly detection, and explainability using LLMs.
- **Frequent Feedback Loops:** Regular interaction with end-users and stakeholders ensures that insights and feedback are incorporated into each sprint cycle, resulting in a more user-aligned solution.
- **Modular Integration:** Machine learning models (e.g., Autoencoder,

Isolation Forest, LOF, One-Class SVM) are developed and validated in independent modules, making them easier to test, refine, and deploy.

- **Continuous Improvement:** Each iteration includes:
 - New data collection from live file operations
 - Feature re-engineering and model retraining
 - Feedback-driven enhancements to system UI and functionality
 - Integration of updated models into the backend

3.6.2 Benefits of Agile for This Project

- Enables rapid adaptation to evolving user needs and data patterns
- Supports ongoing optimization of anomaly detection performance
- Aligns well with the feedback loop required for ML model explainability
- Facilitates real-world testing and validation of detection accuracy

By following Agile, the final system will be both robust and responsive, capable of adapting to changing file behaviors and threat landscapes in real-time.

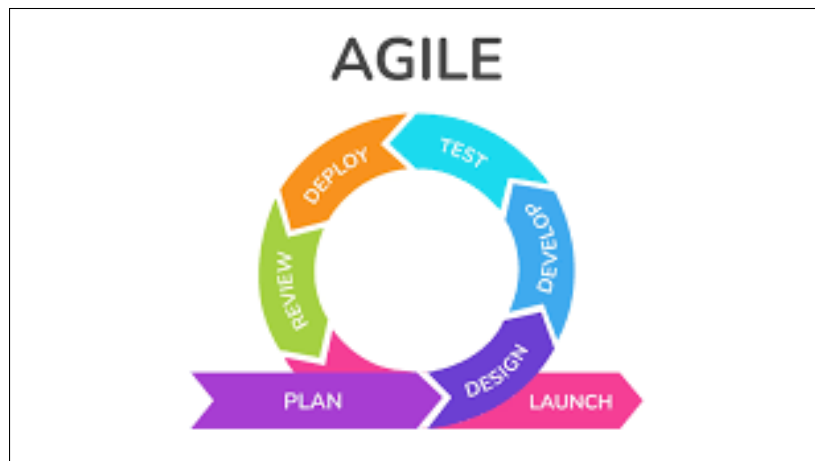


Figure 3.1: Agile model

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

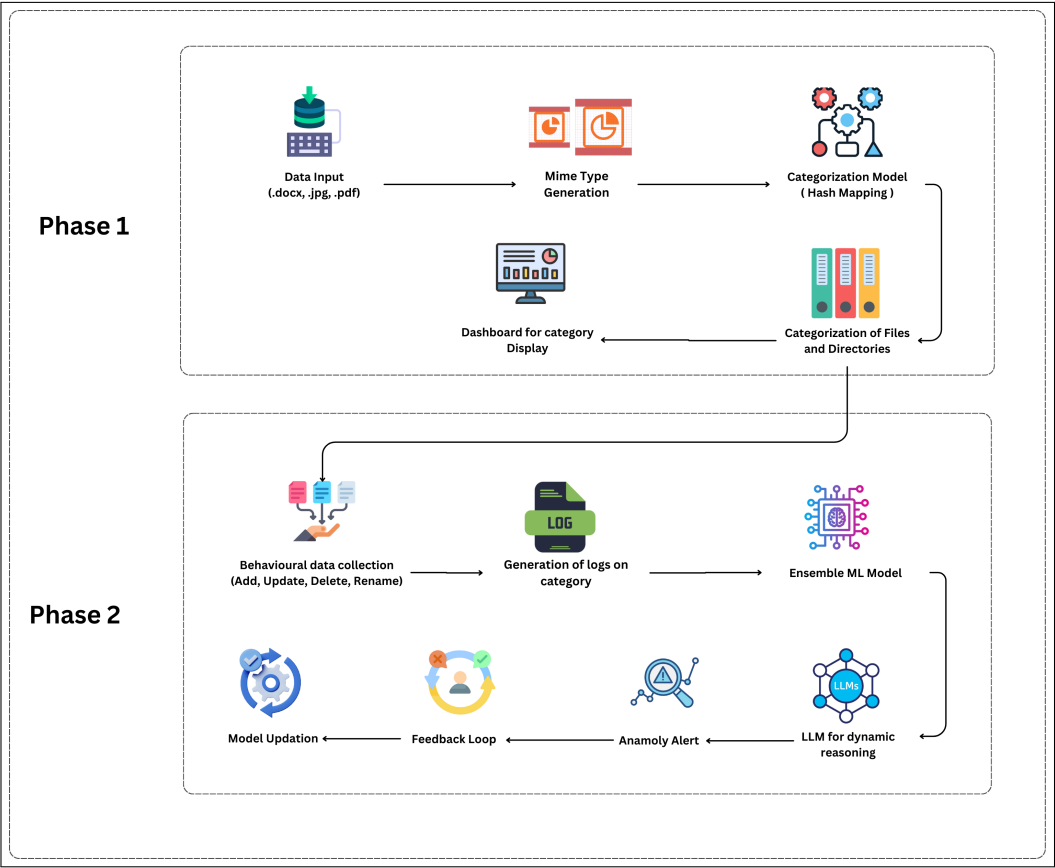


Figure 4.1: Architecture diagram

4.2 UML Diagrams

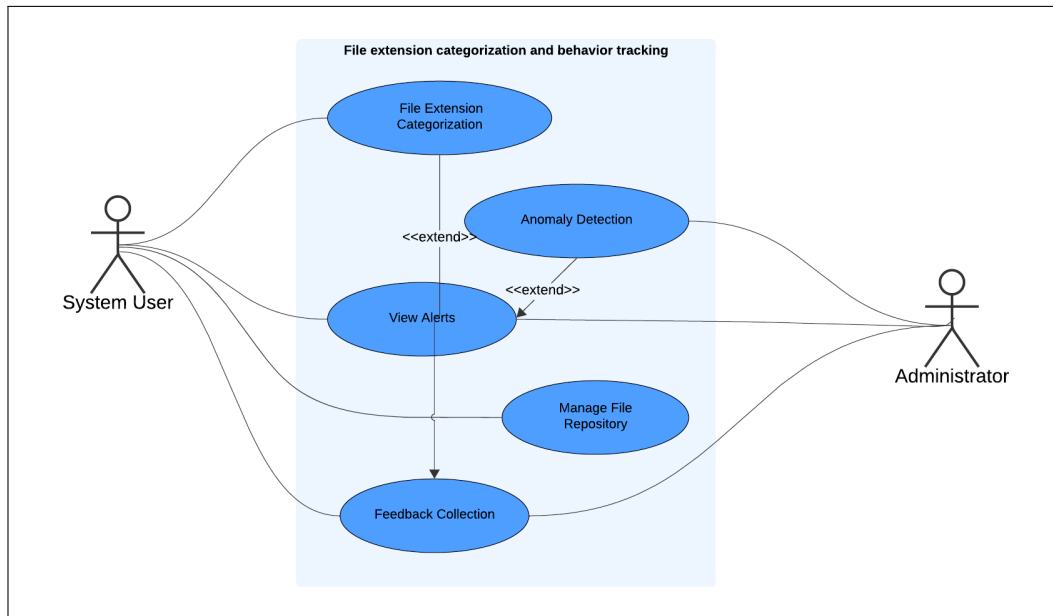


Figure 4.2: Use Case Diagram

4.3 Entity Relationship Diagram

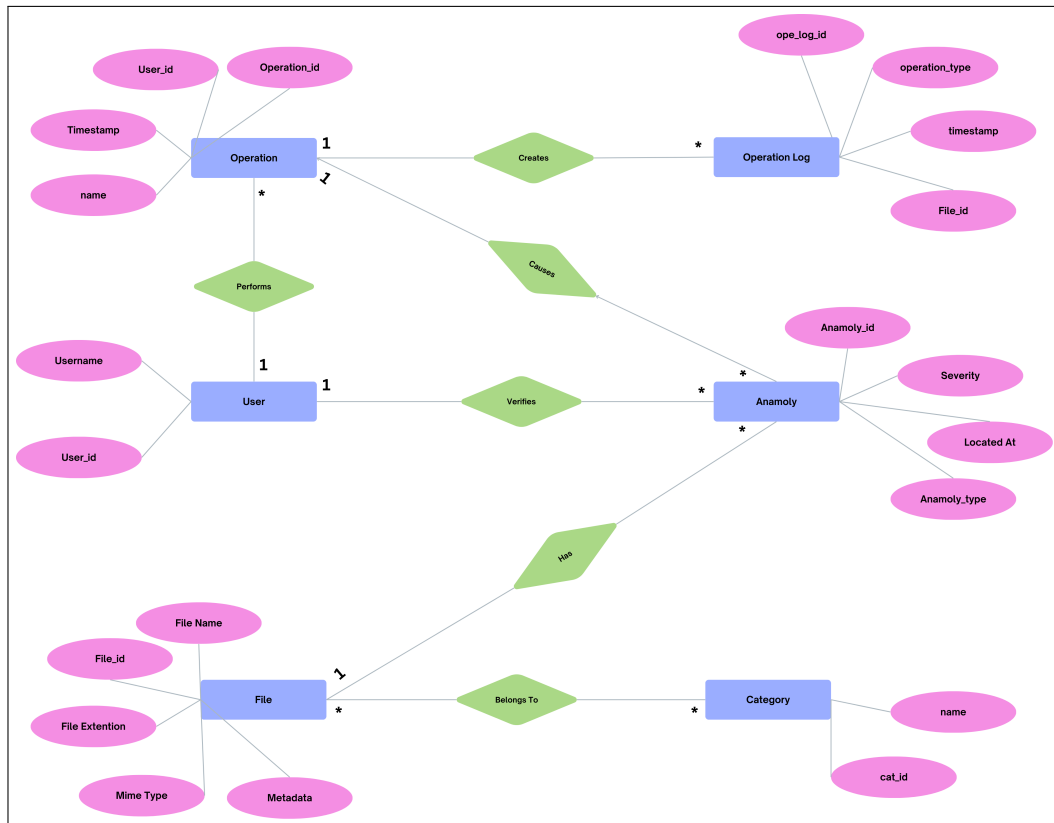


Figure 4.3: ER diagram

4.4 Flow Chart Diagram

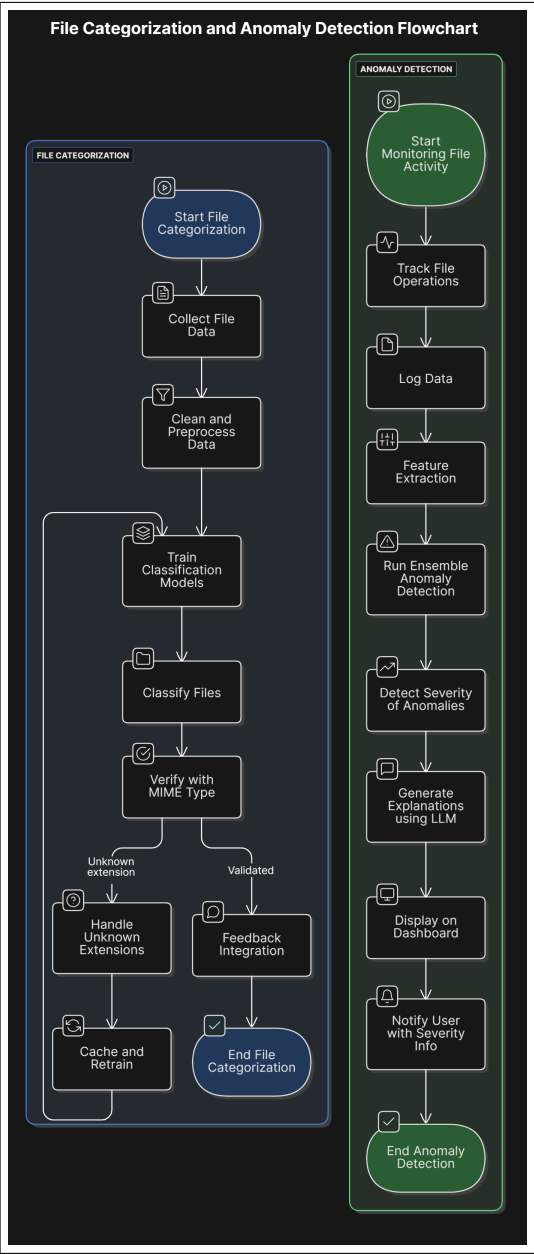


Figure 4.4: Flow Chart Diagram

CHAPTER 5

PROJECT PLAN

5.1 Project Estimate

5.1.1 Reconciled Estimates

After further assessment and analysis, the project timeline and resource estimates have been revised to account for unforeseen challenges and the need for additional refinement in certain areas. The reconciled estimates are as follows:

- **Data Collection and Preprocessing (6 weeks):** An additional week has been allocated for data collection and preprocessing due to the complexity of gathering file extension data and historical logs from various publicly available sources. Additional efforts will be required to handle diverse file types and preprocess them effectively for model training.
- **Model Development and Training (10 weeks):** The model development phase has been extended to 10 weeks to accommodate more comprehensive testing and validation. The complexity of fine-tuning multiple models, particularly for anomaly detection, has necessitated more time for optimizing hyperparameters and ensuring that both models are well-suited for the project goals.
- **Testing, Feedback Incorporation, and Refinement (4 weeks):** The testing phase has been shortened to 4 weeks. However, more frequent feedback loops will be incorporated during model development and training to allow for earlier validation and refinement, which will mitigate delays in the final testing phase.

Total Duration: 20 weeks → 20 weeks (No change in overall timeline)

- The overall timeline remains the same as initially estimated, but additional weeks have been redistributed among the phases to address the project's complexity.

5.2 Risk Management

5.2.1 Risk Identification

Several potential risks could impact the project:

- **Data Collection Delays:** Gathering sufficient and diverse file extension and file operation datasets might take longer than anticipated.
- **Model Complexity:** The chosen machine learning algorithms may prove inadequate for handling edge cases or certain complex file behaviors, necessitating model adjustments or the use of hybrid models.
- **Inaccurate Feedback:** User feedback may not always accurately reflect the system's performance, affecting the quality of retraining efforts.

5.2.2 Risk Analysis

- **Data Collection Delays:** This is a moderate risk as there are several publicly available datasets, but the specific granularity required (e.g., rare file extensions) may pose challenges.
- **Model Complexity:** High risk due to the possibility of the model failing to generalize across different file types or misclassifying rare file extensions.
- **Inaccurate Feedback:** This risk is moderate but manageable with clear instructions to users and administrators on how to provide actionable feedback.

5.2.3 Overview of Risk Mitigation, Monitoring, Management

Risk mitigation will involve continuously monitoring progress at each project stage and employing fallback strategies. For example:

- If data collection faces delays, synthetic data may be generated for initial model training.
- Complex model behavior will be addressed through iterative development cycles, utilizing simpler models (e.g., decision trees) first before scaling to more complex algorithms (e.g., deep learning).
- Regular feedback sessions will be held with users to ensure feedback is constructive and leads to measurable model improvements.

5.3 Project Schedule

5.3.1 Project Task Set

- **Phase 1: Data Collection**
 - Collect datasets of file extensions from authoritative sources (e.g., fileinfo.com)
 - Acquire historical file operation logs from monitored systems
 - Capture metadata such as MIME type, category, and description
- **Phase 2: Data Preprocessing**
 - Clean, normalize, and deduplicate collected datasets
 - Format datasets to match model input specifications
 - Perform categorical encoding and feature extraction
- **Phase 3: Model Development and Training**
 - Develop classification models (Decision Tree, Random Forest, Hash Mapping)
 - Implement multiple anomaly detection models (Isolation Forest, LOF, One-Class SVM, Autoencoder, River)
 - Build ensemble model for robust anomaly detection
 - Conduct hyperparameter tuning and optimization
- **Phase 4: System Integration and Architecture**
 - Integrate ML models into the Flask backend with ZODB storage
 - Create logging system using OS monitoring tools (e.g., watchdog)
 - Deploy LLM-based explanation engine (LLaMA3 via Ollama or Groq)
 - Design API endpoints for model communication
- **Phase 5: Testing and Evaluation**
 - Validate models on unseen datasets for accuracy and generalizability
 - Measure performance metrics (precision, recall, F1-score, inference time)

- Assess quality and consistency of LLM-generated explanations
- Gather user feedback and apply necessary improvements
- **Phase 6: Visualization and Deployment**
 - Build a dynamic React-based dashboard using ReCharts
 - Visualize anomalies, file operations, and categorization trends
 - Implement alert notifications and export functionalities
 - Prepare system for deployment in monitoring environments
- **Phase 7: Documentation and Final Presentation**
 - Document software architecture, design decisions, and user manuals
 - Finalize academic report, presentation slides, and demo videos
 - Prepare for project evaluation, seminar, and defense

5.3.2 Task Network

A task network is designed to visualize dependencies among tasks. For example, the anomaly detection model cannot be developed until the file operation data is collected and preprocessed. Testing and feedback incorporation follow after model development.

5.3.3 Timeline Chart

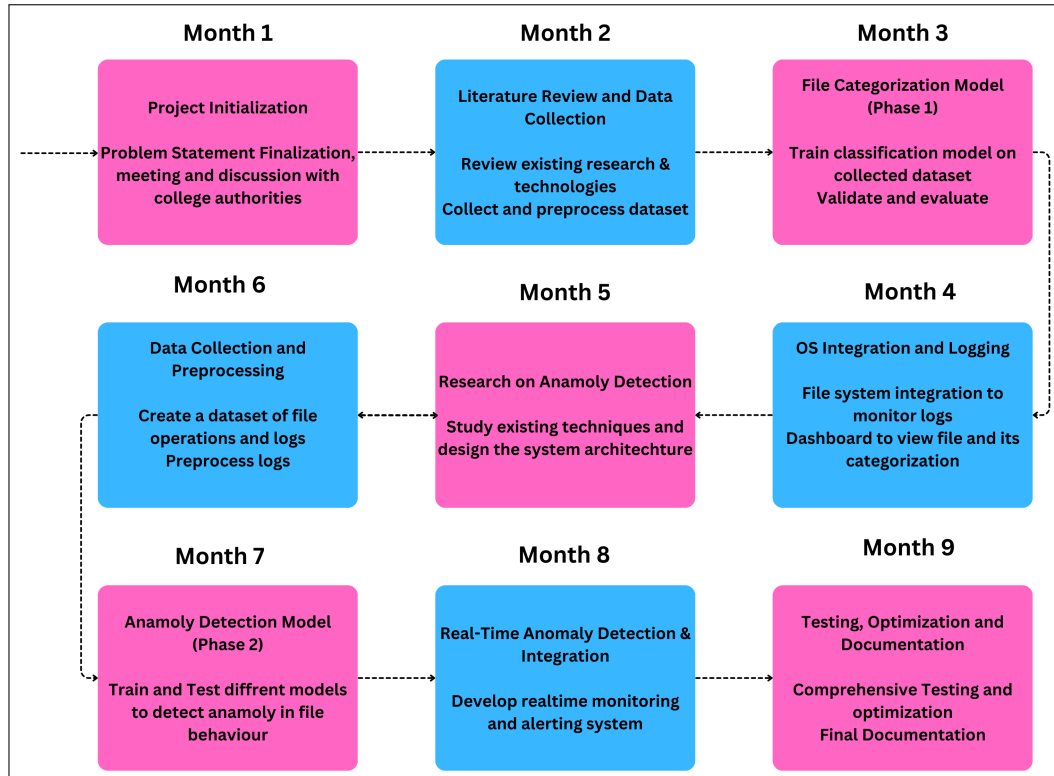


Figure 5.1: Timeline Chart

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

The proposed system is developed using a modular and scalable architecture that follows a phase-wise implementation:

- **File Categorization (Phase 1)**
- **Anomaly Detection (Phase 2)**
- **Explainable Anomalies (Phase 3)**

Each phase builds on the previous one, integrating real-time monitoring, machine learning, and explainable AI to create a robust and intelligent file management and security system.

6.1.1 Phase 1: File Categorization

This phase focuses on categorizing files based on both metadata and content using machine learning and rule-based logic.

Module 1: File Operation Logger This module performs real-time logging of file activities and collects necessary data for categorization.

Captured Attributes Include:

- File name and extension
- MIME type (verified using the magic library)
- File operation (Insert, Modify, Rename, Delete)
- Timestamp and user/process ID
- File path and size

Module 2: Categorization Engine This module classifies files into categories such as Documents, Images, Media, Code, etc.

Key Features:

- **Supervised ML Models:** Decision Tree and Random Forest trained on 20K+ labeled file extensions and MIME types.
- **Hash Data Structure:** Used for faster and more accurate categorization than ML in known cases.
- **Directory/File-Based Input:** Classification supported at both directory and individual file levels.

- **Dynamic Learning:** Unknown extensions are cached and added to the dataset once a threshold is reached, triggering retraining.
- **Feedback Loop:** Incorrect classifications are flagged and corrected by users, enhancing model accuracy over time.

6.1.2 Phase 2: Anomaly Detection

This phase is responsible for identifying abnormal file behavior through continuous monitoring and behavioral modeling.

Module 3: Anomaly Detection Engine This module monitors operations and detects deviations using advanced ML techniques.

Core Functionalities:

- **Feature Extraction:** Extracts meaningful features like operation frequency, time gaps, file sizes, and behavioral trends.
- **Model Stack (Ensemble):**
 - Isolation Forest
 - Local Outlier Factor
 - One-Class SVM
 - Autoencoder
 - River model for streaming data
- **Severity Classification:** Anomalies are tagged with severity levels (low, medium, high) based on feature deviation.
- **Offline and Parallel Execution:** Optimized for large-scale logs with multi-threaded batch processing.

6.1.3 Phase 3: Explainable Anomalies

This phase adds interpretability to the anomalies detected, making the system transparent and actionable.

Module 4: Explainability Engine Components:

- **Clustering:** DBSCAN clusters similar anomalies to reduce redundancy and improve explanation efficiency.
- **LLM Explanation:** Uses Ollama-hosted Llama3 to generate natural language descriptions of why an anomaly was flagged.

- **Explanation Storage:** All LLM outputs are stored in a CSV format and displayed in the frontend dashboard.

Module 5: Dashboard & Reporting

- **Real-Time Alerting:** Notifies users/admins about high-severity anomalies.
- **Interactive Dashboards:** Visual representation of categories, anomalies, and system trends.
- **Reports:** Weekly/monthly summary reports with anomaly trends and model metrics.

6.2 Tools and Technologies Used

The development and deployment of the proposed system leveraged a combination of modern tools, libraries, and platforms to ensure scalability, performance, and explainability. The following technologies were instrumental in different phases of the project:

6.2.1 Programming Language

- **Python 3.11:** Python was the primary language for building all modules due to its simplicity, vast ecosystem of machine learning libraries, and strong community support.

6.2.2 Machine Learning & Deep Learning Frameworks

- **Scikit-learn:** Used for implementing classical ML algorithms like Decision Trees, Random Forests, One-Class SVM, and Local Outlier Factor (LOF).
- **TensorFlow 2.x + Keras:** Used for training deep learning models such as CNNs (for content-based classification) and LSTM (for time-series anomaly patterns).
- **River:** A library for real-time machine learning on streaming data, used in the anomaly detection ensemble.

6.2.3 Explainability & Large Language Models (LLMs)

- **Ollama - Llama3 Model:** Employed for generating natural language explanations of both classification outputs and detected anomalies. Enhances system transparency and helps users understand model behavior.

6.2.4 Monitoring and File System Interaction

- **Watchdog:** A Python library that continuously monitors file system changes (creation, deletion, renaming, modification). It was used to implement the File Logger module.

6.2.5 Data Handling and Processing

- **Pandas:** Used for data cleaning, manipulation, and transformation of log data into model-ready formats.
- **NumPy:** Essential for numerical operations, especially in feature engineering and matrix operations.

6.2.6 Visualization and Reporting

- **Matplotlib & Seaborn:** Used for generating plots, behavior trend graphs, and anomaly heatmaps.
- **ReCharts:** Employed for interactive dashboards and visual reports.

6.2.7 Version Control

- **Git & GitHub:** Used for source code management, version control, and team collaboration.

6.3 Algorithm Details

Input:

- Files present in the system
- Real-time file operations (insert, delete, update, rename)
- Log directory path (C:/Users/<User_Name>/Documents/DriveLogs/logs.fs)

Output:

- Anomalies detected in file operations
- Human-readable explanations for anomalies
- Visualized and filterable dashboard output

6.3.1 Step 1: File Categorization

1. **For each file in the system:**
 - Retrieve its file extension.
 - Verify its actual content type using magic (libmagic) to extract MIME type.
2. **Map files to categories:** (Image, Audio, Video, PDF, Source Code, etc.) using a predefined hashmap.
3. **Store categorized metadata for future validation and feature enrichment.**

6.3.2 Step 2: Real-Time File Logging

1. **Start a persistent file monitoring service using Watchdog.**
2. **For each operation:**
 - Capture:
 - Timestamp
 - Filename & Path
 - File Category (from Step 1)
 - Operation Type (Insert, Update, Delete, Rename)
 - Bytes Modified (for updates)
 - New Filename (for renames)
3. **Serialize and store logs in ZODB (.fs format) for efficient, persistent, and offline-supported storage.**

6.3.3 Step 3: Trigger Anomaly Detection Pipeline

1. **The pipeline is manually triggered by the user through the frontend dashboard (via "Run Pipeline" button).**
2. **The backend loads logs from the configured .fs path in ZODB.**

6.3.4 Step 4: Feature Engineering & Preprocessing

1. **Extract and engineer the following features from logs:**
 - Numerical Timestamp (transformed from datetime)
 - File Category (categorical)
 - Operation Type (one-hot encoded)
 - Bytes Modified (numeric)
 - Operation Frequency (number of similar operations within a time window)
2. **Normalize features where required for consistency across models.**

6.3.5 Step 5: Ensemble Anomaly Detection

1. **Feed preprocessed feature vectors into the following models:**
 - **Isolation Forest:** Detects anomalies based on data isolation
 - **One-Class SVM:** Learns decision boundary for normal ops
 - **Local Outlier Factor (LOF):** Measures local density deviation
 - **Autoencoder:** Deep neural network to reconstruct patterns and detect deviations
 - **River Model:** Incremental stream learning for online anomaly detection
2. **Ensemble logic:**
 - Combine outputs via voting, score averaging, or thresholding.
 - Flag a log entry as anomalous if it crosses the combined threshold.

6.3.6 Step 6: Explainability via Local LLM

1. **For each detected anomaly:** Pass log details, features, and context into a local LLM instance (e.g., LLaMA 3 hosted on Ollama).
2. **Use a templated prompt to generate a natural language explanation:**

“This file rename operation was flagged because it involved a system file being renamed outside of regular hours, which is unusual for this user.”

3. **Store explanation alongside the anomaly record as a key-value pair in JSON or CSV.**

6.3.7 Step 7: Visualization & Dashboard Output

1. **Final structured anomalies (with explanations) are sent to the React frontend dashboard via Flask API.**
2. **Enable advanced filters:**
 - Time Window
 - File Category
 - Operation Type
 - Severity Level / Risk Score
3. **Allow data export in CSV or JSON format.**
4. **Generate rich visualizations:**
 - Time-Series Graphs showing anomaly trends over time
 - Heatmaps of file activity and risk intensity
 - Bar/Pie Charts breaking down anomalies by category or type

CHAPTER 7

SOFTWARE TESTING

7.1 Types of Testing

Thorough testing was conducted throughout the development lifecycle to ensure the robustness, reliability, and usability of the anomaly detection system. Various testing methodologies were applied at different stages, focusing on both individual modules and the complete pipeline.

7.1.1 Unit Testing

Each module was tested independently to validate its core functionality:

- **File Categorizer:** Tested for accurate classification using MIME type and file extensions across a wide range of file formats (e.g., images, documents, code files).
- **Logger Module:** Verified correct logging of operations (insert, delete, update, rename) with appropriate metadata such as timestamps, bytes modified, etc.
- **Storage Layer (ZODB):** Checked for consistency and persistence of logs in the .fs format.
- **Anomaly Detection Models:** Ensured that Isolation Forest, SVM, LOF, Autoencoder, and River-based models were functioning as expected on synthetic and real datasets.
- **LLM Explainability Engine:** Validated that explanations were generated consistently and aligned with the context of anomalies.

7.1.2 Integration Testing

Integration tests were carried out to ensure smooth data flow and interaction between modules:

- Logs generated by the logger were correctly consumed by the anomaly detection pipeline.
- Anomalies detected were seamlessly passed to the LLM engine for explanation.
- Outputs were successfully rendered in the dashboard and matched user expectations.

7.1.3 Functional Testing

All key features of the system were tested against their functional requirements:

- Accurate real-time logging of file operations.
- Correct categorization and anomaly flagging.
- LLM-generated explanations appended to anomaly outputs.
- Filtering, exporting, and visualizing anomalies via the dashboard.

7.1.4 Performance Testing

Stress testing was conducted to evaluate the system's behavior under load:

- Simulated high-frequency file operations to test logging throughput.
- Assessed the response time and efficiency of the anomaly detection pipeline with large log datasets.
- Tested the multi-threaded LLM explanation engine for concurrent processing of multiple anomalies.

7.1.5 User Acceptance Testing (UAT)

End-user scenarios were simulated to evaluate the system from a usability perspective:

- Normal and abnormal file behaviors were tested to check detection accuracy.
- Explanations were reviewed by users to ensure clarity and relevance.
- Dashboard interactivity and responsiveness were evaluated for usability.

7.2 Test cases & Test Results

File Categorization and Anomaly Detection in File Operation Behaviour

Testcase ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-001	File Logger	Create, rename, delete a file	File operation logged with correct metadata	Logs captured correctly	Pass
TC-002	Preprocessing Module	Feed in noisy, duplicate, or incomplete logs	Cleaned data with extracted features	Features extracted	Pass
TC-003	Categorization Model	Input file with .jpg extension	Classified as "Image File"	Correct category	Pass
TC-004	Categorization Model	Input file with unknown extension .xyz	Classified as "Unknown" or "Other"	Handled as "Other"	Pass
TC-005	Anomaly Detection Engine	Simulate a bulk delete of image files	Anomaly detected and alert generated	Alert generated	Pass
TC-006	Anomaly Detection Engine	Feed in normal operation patterns	No anomaly detected	No alert triggered	Pass
TC-007	Alert & Report Generator	Trigger classification and anomaly events	Summary report generated with categorized logs and anomaly stats	Report generated	Pass
TC-008	Edge Case Handling	Insert file with missing metadata (no extension, no MIME type)	Skipped or flagged as invalid input	Handled gracefully	Pass
TC-009	Performance Under Load	Process 10,000 logs within 60 seconds	System handles input without crash or lag	Handled smoothly	Pass
TC-010	Explainability (LLM Output)	Trigger anomaly and check LLM-generated explanation	Clear human-readable description of the anomaly cause	Explanation generated	Pass

Table 7.1: Test Case Results

CHAPTER 8

RESULTS

8.1 Outcomes

The proposed system has been comprehensively evaluated across four core components—file categorization, anomaly detection, LLM-based explanation, and real-time visualization. The evaluation highlights the robustness, accuracy, and practicality of the implemented solution in a real-world file operation monitoring environment.

8.1.1 File Categorization Engine

The intelligent file categorization component combines metadata analysis, supervised machine learning, and MIME-type-based hashing for accurate classification. It supports over 20,000 unique file extensions and more than 400 defined categories.

Table 8.1: Performance Evaluation of File Categorization Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Decision Tree	89.3	88.7	87.9	88.3
Random Forest	93.6	94.1	92.8	93.4
Hash-Based Mapping	99.0	99.0	99.0	99.0

The hash-based MIME mapping provides the highest accuracy with minimal overhead and is adopted as the primary classification method, whereas machine learning models are used for unknown or dynamically observed file extensions.

8.1.2 Anomaly Detection Engine

This module monitors real-time file operations including insertions, updates, deletions, and renaming. Logs are stored using a persistent object database (ZODB), and multiple anomaly detection algorithms are used in an ensemble configuration to identify suspicious behaviors.

The ensemble method delivers the most balanced performance, outperforming individual models in both accuracy and false positive rate while maintaining reasonable execution time.

Table 8.2: Evaluation of Anomaly Detection Models

Model	Detection Accuracy (%)	False Positive (%)	Pos-Rate	Execution Time (ms/sample)
Isolation Forest	87.5	4.8		2.3
One-Class SVM	84.1	6.2		4.1
Local Outlier Factor	80.6	5.7		1.9
Autoencoder	89.4	3.9		5.5
River Model	78.2	7.4		1.1
Ensemble (Proposed)	93.8	2.1		3.7

8.1.3 Explainability via LLMs

Anomalies identified by the system are contextualized using a locally deployed Large Language Model (LLaMA 3 via Ollama). The model generates human-readable explanations for each flagged event, enhancing interpretability and user trust.

Table 8.3: Evaluation of LLM-Generated Explanations

Evaluation Metric	Average Score (Out of 5)
Explanation Clarity	4.7
Relevance to Operational Context	4.6
Usefulness for Non-Technical Users	4.8
Consistency Across Cases	4.5
Inference Time (per sample)	~1.2 seconds

These explanations have proven highly effective for both technical audits and user-facing insights, especially in environments lacking detailed logs or domain expertise.

8.1.4 Real-Time Visualization and Alert System

The system includes an interactive dashboard built using React and ReCharts. It provides real-time monitoring, categorization analytics, anomaly visualization, and explainability integration. Key features include:

- Time window filtering and event aggregation
- Operation-wise breakdown (Insert, Update, Delete, Rename)
- File category and severity-level filtering
- Export options in CSV and JSON formats
- Visualizations: time series graphs, heatmaps, bar charts

The dashboard has been benchmarked for responsiveness and usability, confirming its suitability for continuous monitoring environments.

8.2 Screen Shots

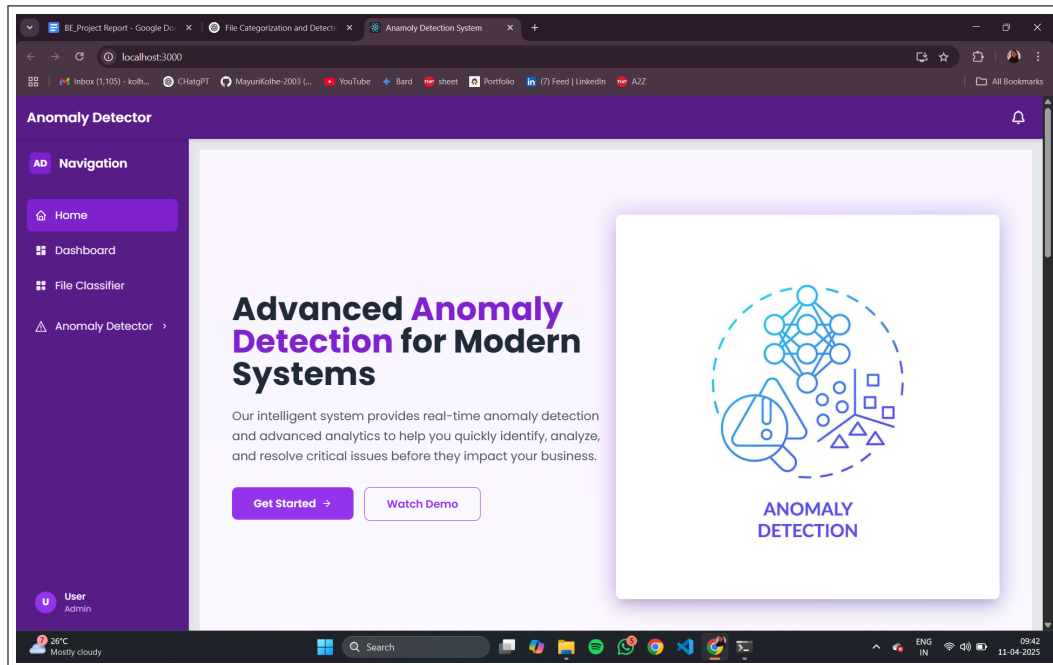


Figure 8.1: Home Page

File Categorization and Anomaly Detection in File Operation Behaviour

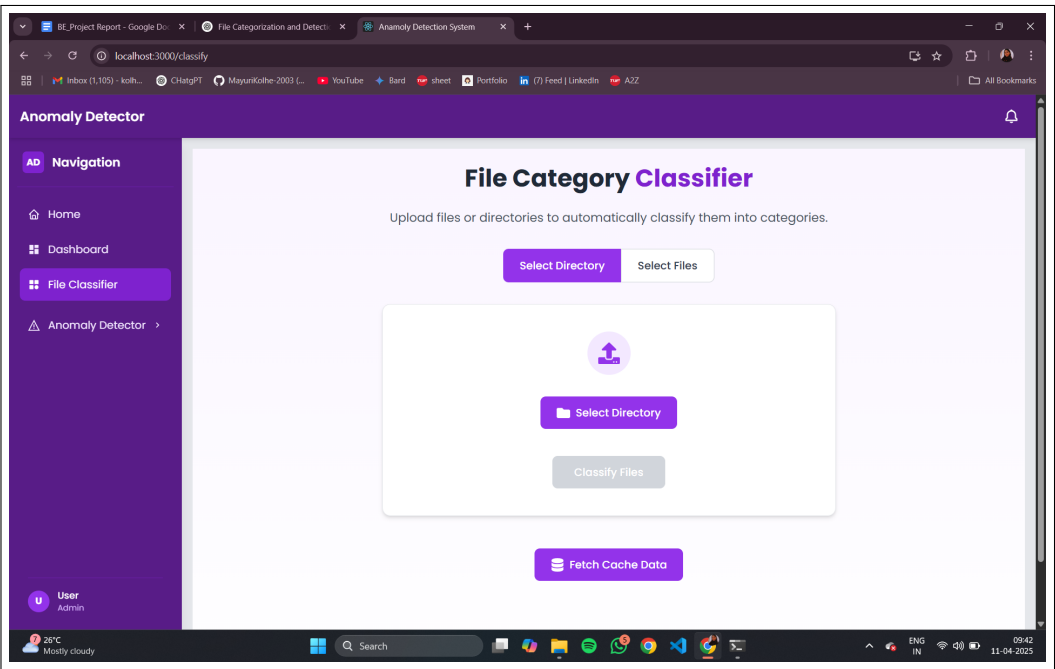


Figure 8.2: File categorizer

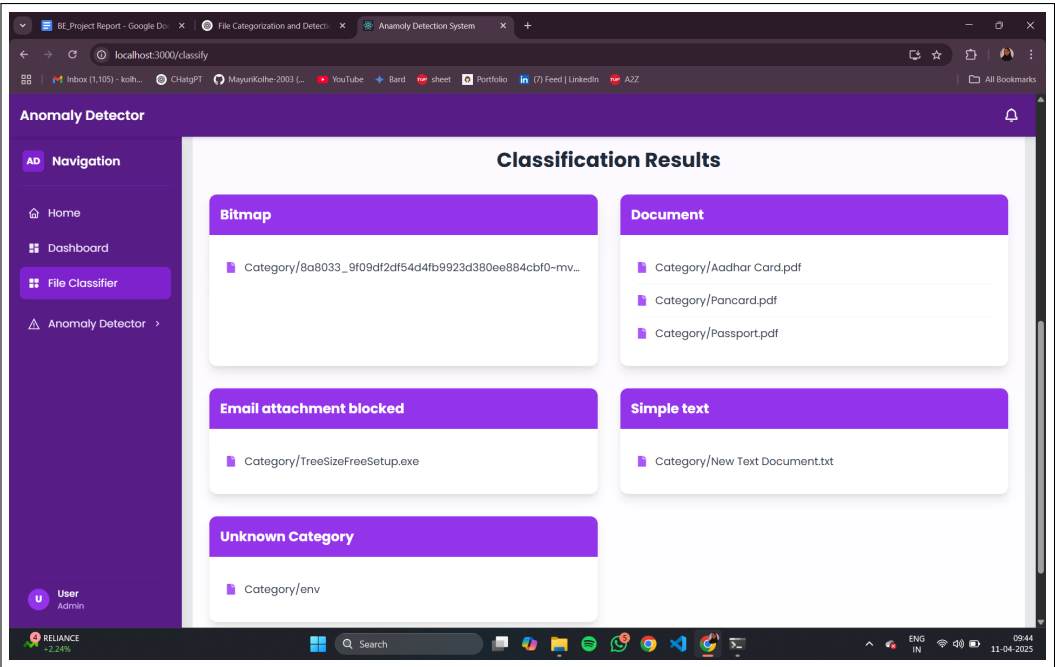


Figure 8.3: File categorizer results

File Categorization and Anomaly Detection in File Operation Behaviour

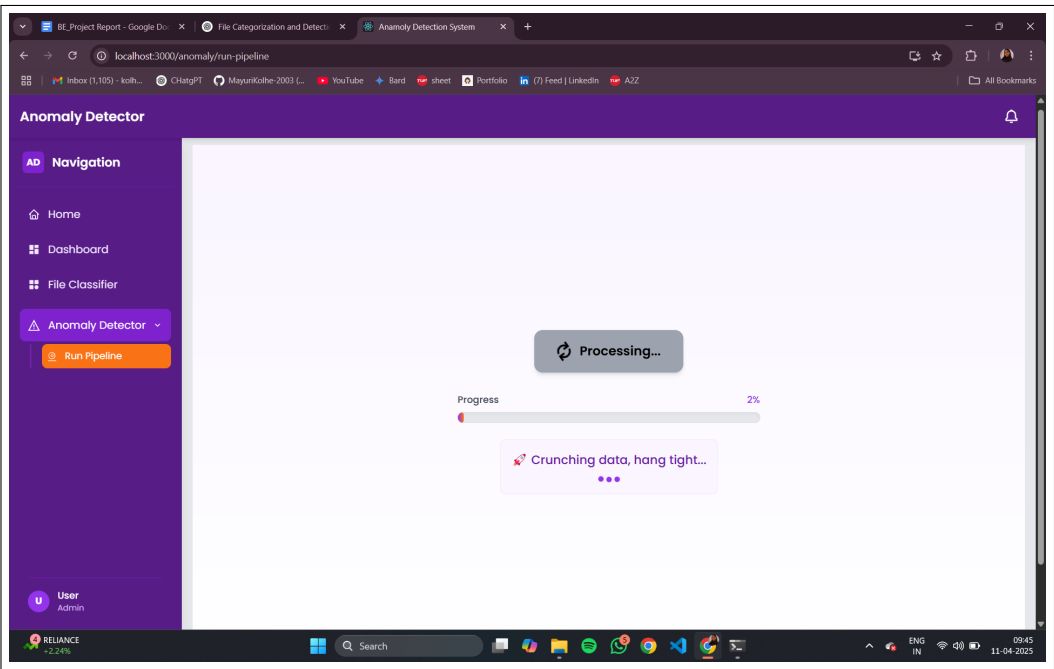


Figure 8.4: Pipeline to run Anomaly Detection

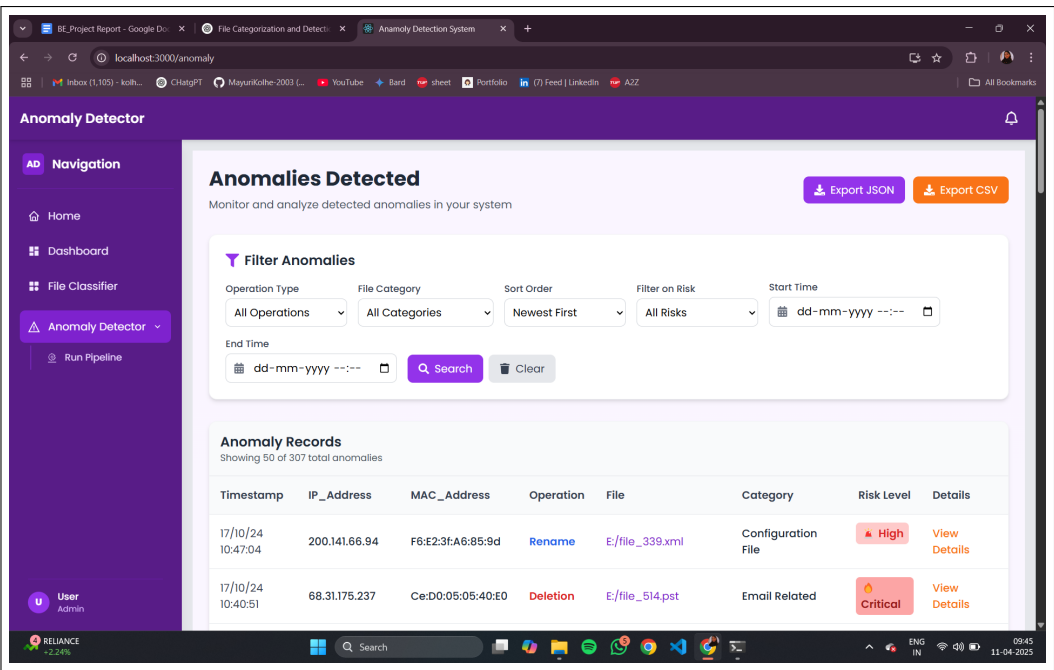


Figure 8.5: Detected Anomalies Dashboard

File Categorization and Anomaly Detection in File Operation Behaviour

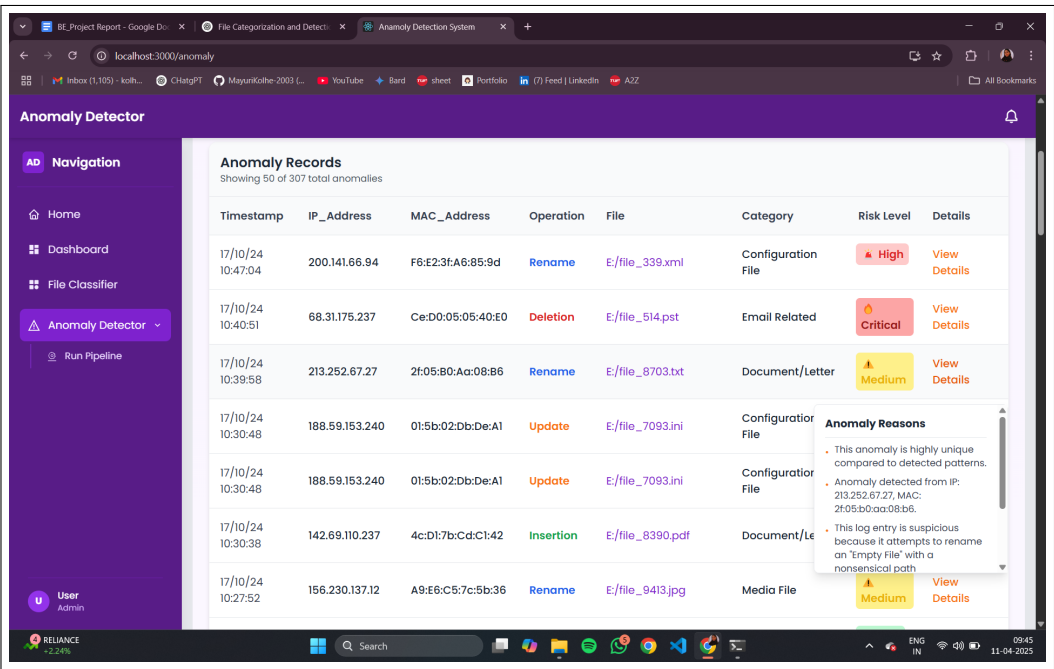


Figure 8.6: Detected Anomalies with reasons

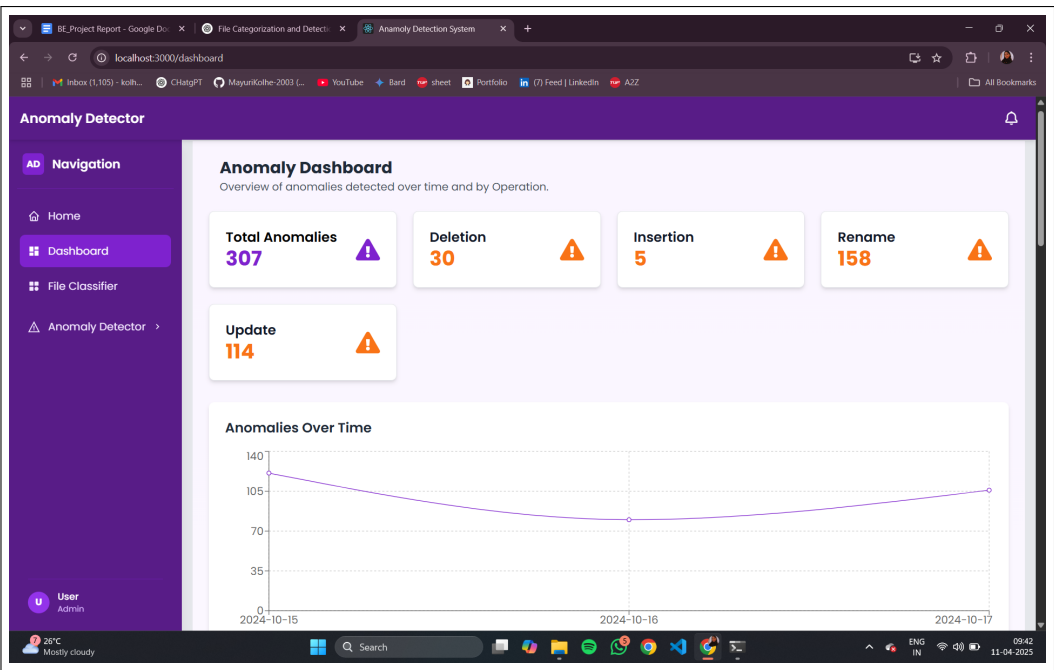


Figure 8.7: Dashboard for Analysis

File Categorization and Anomaly Detection in File Operation Behaviour

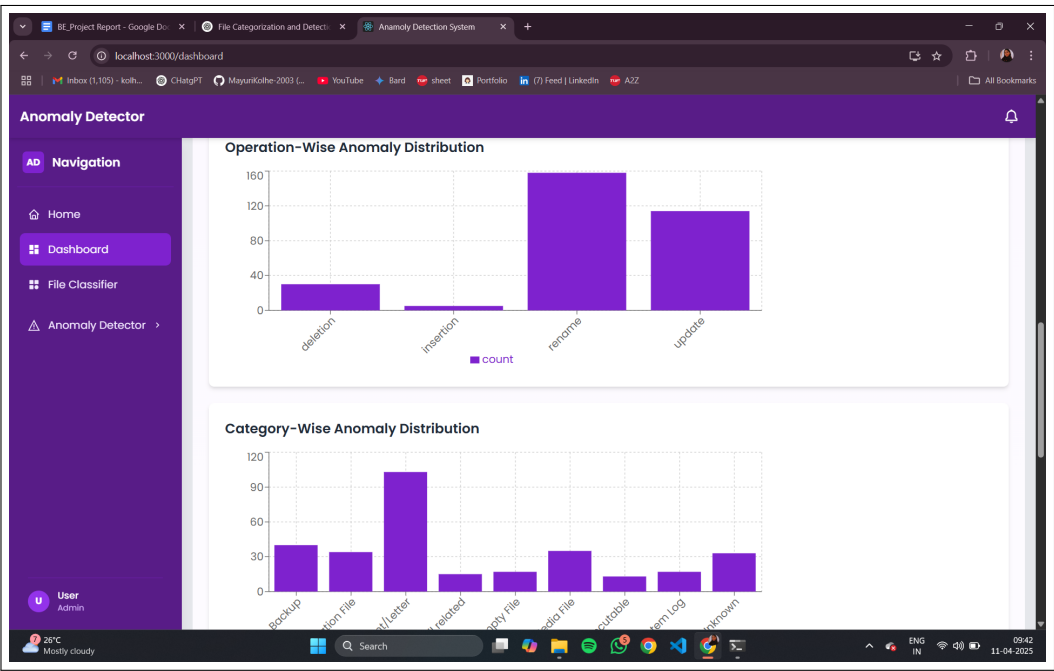


Figure 8.8: Anomaly distribution over operation and categories

CHAPTER 9

CONCLUSIONS

9.1 Conclusions

The project successfully delivers a comprehensive solution for automated file categorization and intelligent anomaly detection within file systems. By leveraging a combination of supervised and unsupervised machine learning techniques, the system accurately classifies files based on their extensions and content, while simultaneously monitoring behavior patterns to detect suspicious or abnormal activities in real-time. The integration of explainable AI models enhances user trust by providing clear, human-readable justifications for both classification and anomaly alerts. The real-time dashboard further contributes to the system's usability, offering intuitive visualization and actionable insights. Overall, the system not only improves file organization and operational efficiency but also significantly strengthens data security and governance, making it well-suited for deployment in enterprise-level and security-critical environments.

9.2 Future Work

The current system presents a robust foundation for intelligent file categorization, anomaly detection, and explainable monitoring. However, several enhancements can be pursued to improve adaptability, scalability, and system intelligence. Future development directions include:

9.2.1 User Behavior Profiling

To further improve anomaly detection precision, future work will involve modeling user- or role-specific behavioral baselines. By analyzing historical file operation patterns for individual users or departments, personalized anomaly thresholds can be defined, significantly improving the system's ability to detect insider threats or targeted malicious activity.

9.2.2 Real-Time Remediation Mechanisms

Future versions of the system may integrate automated response mechanisms that trigger remediation actions in real-time. For example:

- Blocking unauthorized file operations
- Quarantining suspicious files
- Triggering role-based access control checks

- Sending prioritized alerts to system administrators

This will transition the system from a passive monitoring tool to an active defense mechanism.

9.2.3 Continuous and Online Learning Models

Current models require periodic retraining to accommodate evolving file behaviors or new file types. Future iterations will integrate online learning and adaptive algorithms that incrementally update themselves based on new data streams, minimizing system downtime and reducing retraining overhead.

9.2.4 Multi-Platform and Cloud Integration

To expand the system's applicability, support for monitoring cloud-based file systems will be incorporated. This includes integration with platforms such as:

- Google Drive
- Dropbox
- Microsoft OneDrive
- AWS S3

Such extensions will allow unified anomaly detection across hybrid environments (local + cloud) and ensure seamless categorization and monitoring irrespective of storage location.

9.2.5 Enhanced LLM Personalization and Fine-Tuning

Future improvements to the LLM-powered explanation module include persona-aware fine-tuning and domain adaptation. This will enable explanations tailored to specific user roles (e.g., system admins vs. auditors) and sectors (e.g., education, healthcare, corporate), thereby improving interpretability and relevance.

9.3 Applications

9.3.1 Enterprise File Management Systems

Automates file organization and monitors unusual activities in corporate environments, helping IT teams manage data efficiently and detect internal threats.

9.3.2 Cybersecurity and Threat Detection

Identifies potential security breaches through behavioral anomalies, such as mass deletions or unauthorized access, aiding in early threat detection and prevention.

9.3.3 Cloud Storage Monitoring

Integrates with cloud platforms to classify uploaded files and monitor user activity, ensuring compliance and security in cloud-based infrastructures.

9.3.4 Digital Forensics and Audit Trails

Provides detailed logs and behavioral analysis that can be used in forensic investigations to trace suspicious operations or policy violations.

9.3.5 Data Loss Prevention (DLP) Systems

Helps organizations detect and respond to risky file operations, preventing accidental or malicious data leaks.

CHAPTER 10

APPENDIX

10.1 Appendix A

Problem statement feasibility assessment using, satisfiability analysis and NP Hard, NP-Complete or P type using modern algebra and relevant mathematical models.

This appendix evaluates the feasibility of the project titled “*A Dual-Phase Framework for Intelligent File Categorization and Real-Time Anomaly Detection in File Operation Behaviors*” by applying logical satisfiability, computational complexity theory, and modern algebraic modeling techniques.

1. Satisfiability Analysis

Phase 1 – File Categorization:

The problem is modeled as a deterministic classification task using file extensions and MIME types. For standard file types, the task is satisfiable and efficiently solvable using hash-based lookups and supervised models. Challenges arise with ambiguous or previously unseen extensions, which may require dynamic learning strategies.

Phase 2 – Anomaly Detection:

The task of identifying behavioral anomalies in file operations (e.g., mass deletions, renames) involves handling real-time data with inherent variability. While detection is satisfiable in small-scale scenarios, the problem becomes increasingly complex and less predictable at scale due to asynchronous events and overlapping operations.

2. Computational Complexity Classification

array

3. Application of Modern Algebra

Group Theory:

Applied to model categorization operations as transformations on file-type sets, enabling structured classification logic with reversibility and consistency.

Graph Theory:

Used in anomaly detection to represent file behavior as a directed graph where:

Task	Complexity Class	Justification
Basic File Categorization (Extension-based)	P (Polynomial)	Operates on hash mappings and decision trees; completes in deterministic time.
Handling Ambiguous/Mislabeled Files	NP-Complete	Requires combinatorial resolution and probabilistic inference.
Real-Time Anomaly Detection	NP-Hard	Involves dynamic data streams, multivariate analysis, and real-time constraints.

Table 10.1: Complexity Classification of Project Tasks

- Nodes denote file states (e.g., created, modified, renamed).
- Edges represent operations, allowing traversal and anomaly scoring based on unusual patterns or cycles.

4. Conclusion

The proposed dual-phase system is partially feasible within defined boundaries:

- **File Categorization** is computationally tractable and robust using traditional algorithms and hash structures.
- **Anomaly Detection** introduces scalability and performance challenges due to its NP-Hard nature in real-time, dynamic environments.

The use of algebraic modeling adds formal structure, improving the interpretability and manageability of system behaviors. This analysis supports the viability of the framework, provided that optimizations, heuristics, and scalable architectures are employed for the anomaly detection phase.

10.2 Appendix B

Details of paper publication: name of the conference/journal, comments of reviewers, certificate, paper.

- **Paper Title:** *Enhancing File Management Systems with AI: A Dual-Phase Approach to Categorization and Anomaly Detection*
- **Conference Name:** 9th International Conference on Control Communication, Computing and Automation (ICCUBEA 2025)
- **Track Name:** Information, Cyber and Network Security
- **Paper ID:** 338
- **Date of Submission:** March 31, 2025

10.3 Appendix C

Plagiarism Report of project report.

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CHAPTER 11

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